



Object Oriented Programming

Pass Task 6.2: Key Object Oriented Concepts

Overview

You have been using object oriented programming to implement the programs you have created in this unit. In this task you need to express your understanding of the principles associated with this programming paradigm and how they relate to one another.

Purpose: Express your understanding of how the programming artefacts, tools, and concepts relate to each other.

Task: Write a short explanation of object oriented programming to outline your understanding of the associated principles and tools, and how they are used to create programs. Create an accompanying concept map to show how the concepts are linked.

Time: **End of Week 7, Friday, 20 September 2024, 23:59:00 (Firmed)**

Submission Details

All students have access to the Adobe Acrobat tools. Please print your solution to PDF and combine it with the screenshots taken for this task.

- A PDF document with your description and accompanying concept map.

Make sure that your task has the following in your submission:

- For written explanations:
 - Explanation should clearly demonstrate a good understanding of the object oriented programming principles and how they relate to the development of software.
 - The report is your work and expresses your understanding in your own words. Where ideas and descriptions are related to other people they are appropriately cited and referenced.
- For the concept map:
 - Demonstrates thought in what is presented and how it is presented.
 - Relationships are clearly shown and annotated with explanations, not just listed.

Instructions

To make the most of programming, you need to deeply understand the principles that underly the paradigm that you are using.

For this task you must **explain** the four key principles of object oriented programming and **re-late** these to the programs you have created. You must also **visualise** how these principles are and other concepts are related in a concept map.

Note: An explanation is more than just a description, you want to convey a deeper understanding that can be achieved with a simple description. Try to relate the principles together and express the depth of your understanding.

Tip: A deep explanation does not need to be a long one. Aim to convey your understanding in a concise fashion.

Create a document that explains the four key principles of object oriented programming, as you understand them. Draw a concept map to accompany your explanation, that elaborates on the relationships between the four key principles, other concepts, and programming artefacts.

Tip: Including references to textbooks (not wikipedia) or research papers can help you strengthen your explanation. Remember to reference other peoples work.

Requirements:

1. Keep it to about 1 or 2 pages of text
2. Structure the written portion of your document as a list. Each item in the list should clearly communicate: which term you are defining, a definition of the term, and an example.
3. Relate the principles to programs **you** have written in this unit.
4. Once you have written the text portion, draw a concept map showing how these four principles are connected to each other and other concepts and artefacts we have used in the unit.

Tip: In your concept map, only include the most important connections or it will become messy and hard to read.

Make sure your concept map covers the following (as a minimum, feel free to extend this):

Concepts	Artefacts	Action	Terminology
Abstraction	Class	Method Call	Value Type
Encapsulation	Object	New	Reference Type
Inheritance	Interface		Abstract Class
Polymorphism	Method		Abstract Methods
Roles	Fields		Private
Responsibilities			Public
Collaborations			Protected
Coupling			Overload
Cohesion			Override
			Virtual

Provide three appropriate examples where Exception Handling should be used.

Assessment Criteria

Make sure that your task has the following in your submission:

- The four key principles are clearly and accurately defined in text.
- An appropriate example is provided for every definition.
- Explanations are written in your own words and sources are correctly cited.
- Every term listed in the table is covered in the concept map, with appropriate annotated connections.
- The “Universal Task Requirements” (see Canvas) have been met.

The first key principle of OOP is Encapsulation which involves grouping methods and variables into their own groups or classes to allow for a much more simpler objects, you would employ this by restricting the access from outside the class and only using getters and setters to access the data, using them as inputs and outputs for the information, this makes each step of a program more manageable and secure.

The second principle of OOP is Inheritance, this is the method of using sub-classes to inherit variables and methods from other classes by referencing those sub-classes.

The third principle of OOP is Abstraction and this is the process of starting an object at its most basic properties and then using classes and sub-classes to give detail to that object, it removes the process of giving that object those properties and stream lines the process so that the user dose not need to think about how it is made.

The final principle in Polymorphism which is the process of using methods for more then one task depending on the data that is given, this reduces the amount of code needed and saves on memory like in the Shape Drawer, using the Drawing class to draw the shapes but it would change depending on the shape that was selected.

